

# FractionalMaxPool3d#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.pooling.FractionalMaxPool3d.html>

## Description:

Applies a 3D fractional max pooling over an input signal composed of several input planes. Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham The max-pooling operation is applied in  $k_T \times k_H \times k_W$  regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes. Exactly one of `output_size` or `output_ratio` must be defined.

## Function Signature

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```
class
torch.nn.modules.pooling.FractionalMaxPool3d(kernel_size,
output_size=None, output_ratio=None, return_indices=False,
_random_samples=None)[source]#
```

Parameters

Shape:

## Code Examples

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```
>>> # pool of cubic window of size=3, and target output size
13x12x11
>>> m = nn.FractionalMaxPool3d(3, output_size=(13, 12, 11))
>>> # pool of cubic window and target output size being half of
input size
>>> m = nn.FractionalMaxPool3d(3, output_ratio=(0.5, 0.5, 0.5))
>>> input = torch.randn(20, 16, 50, 32, 16)
>>> output = m(input)
```

## Notes & Warnings

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**NOTE** Exactly one of `output_size` or `output_ratio` must be defined.

## Detailed Documentation

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### Documentation

Applies a 3D fractional max pooling over an input signal composed of several input planes.

Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham

The max-pooling operation is applied in  $k_T \times k_H \times k_W$  regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes.

Exactly one of `output_size` or `output_ratio` must be defined.

`kernel_size` (Union[int, tuple[int, int, int]]) – the size of the window to take a max over. Can be a single number  $k$  (for a square kernel of  $k \times k \times k$ ) or a tuple  $(k_t \times k_h \times k_w)$ ,  $k$  must be greater than 0.

`output_size` (Union[int, tuple[int, int, int]]) – the target output size of the image of the form  $o_T \times o_H \times o_W$ . Can be a tuple  $(o_T, o_H, o_W)$  or a single number  $o_H$  for a square image  $o_H \times o_H \times o_H$

`output_ratio` (Union[float, tuple[float, float, float]]) – If one wants to have an output size as a ratio of the input size, this option can be given. This has to be a number or tuple in the range (0, 1)

`return_indices` (bool) – if True, will return the indices along with the outputs. Useful to pass to `nn.MaxUnpool3d()`. Default: False

Input:  $(N, C, T_{in}, H_{in}, W_{in})$  or  $(C, T_{in}, H_{in}, W_{in})$

Output:  $(N, C, T_{out}, H_{out}, W_{out})$  or  $(C, T_{out}, H_{out}, W_{out})$ , where  
 $(T_{out}, H_{out}, W_{out}) = \text{output\_size}(T_{out}, H_{out}, W_{out})$   
or  $(T_{out}, H_{out}, W_{out}) = \text{output\_ratio} \times (T_{in}, H_{in}, W_{in})$

